Carlos Ramirez

Software Engineer | Business Analytics
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Experience

Software Engineering Intern | Lam Research May 2025 – August 2025 | Fremont, CA

- Built a **real-time telemetry ingestion pipeline** in Python and FastAPI that collected sensor data from 20+ semiconductor etch chambers, reducing process anomaly detection latency **from 2 hours to under 5 minutes**
- Developed internal **REST API services** to expose equipment utilization and process health metrics to downstream manufacturing execution systems, processing **50,000+ data events per shift**
- Integrated real-time equipment signals with **PostgreSQL-backed historical trend analysis** to power threshold-based predictive maintenance alerting, in collaboration with process engineers

Data Analyst | The Global Career Accelerator August 2023 – January 2024 |

- Analyzed datasets of **50,000+ records** in Jupyter Notebook to uncover performance trends, delivering insights that improved decision-making speed by 30%
- Built **Tableau dashboards** and visual reports adopted by 5+ cross-functional teams, reducing reporting time from days to hours
- Identified 10+ behavioral patterns enabling leadership to refine product strategies and improve user engagement by 20%; presented to an international team spanning **4 continents** with 99% on-time delivery

Founder & Software Engineer | Pool Map 2025 – Present |

- Built and shipped a cross-platform iOS and web app using **React Native (Expo) and TypeScript**, live on Apple TestFlight, enabling users to discover, add, and rate swimming pools on an interactive map
- Designed a multi-table **PostgreSQL schema** with SQL migrations and a self-hosted **Express.js REST API**, powering a leaderboard ranking system via optimized relational queries

Projects

Serverless Inventory API | *Python, AWS Lambda, DynamoDB, API Gateway, CloudWatch* 2025

- Designed and deployed a cloud-native REST API on **AWS Lambda and API Gateway** with DynamoDB as the primary store, sustaining 10,000+ req/min at sub-100ms p99 latency
- Built a fully automated **CI/CD pipeline with GitHub Actions** and instrumented CloudWatch alarms for proactive monitoring, achieving 99.9% uptime

Intel Corporate Sponsorship | *SQL, Python, Jupyter Notebook, Git* 2024

- Directed SQL-based analysis of **100,000+ records** to identify optimal regions for Intel's new semiconductor data center, scoping findings across energy cost, output, and infrastructure criteria
- Translated complex data findings into **clear business recommendations** for corporate leadership, enabling data-driven site selection decisions across a cross-functional stakeholder group

Education

Arizona State University Tempe, AZ

Master of Science in Business Analytics, W.P. Carey School of Business *Expected May 2027*

Bachelor of Science in Computer Science, Ira A. Fulton Schools of Engineering *May 2026*

Technical Skills

Languages: Python, Java, C++, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS
Frameworks & Libraries: React, Node.js, Express, FastAPI, Pandas, NumPy, TensorFlow
Data & Analytics: Tableau, Jupyter Notebook, Excel, REST APIs
Developer Tools & Cloud: Git, Docker, GitHub Actions, AWS (Lambda, DynamoDB, API Gateway, CloudWatch), GCP
AI & LLM Tools: Claude, ChatGPT, GitHub Copilot, Cursor; prompt engineering, LLM API integration